

# Experiment Report

Iterative Data Selection for Spider Text-to-SQL

Spider 1.0 + CodeT5-small | 21 September 2026

This study investigates whether iterative model-aware data selection can improve Text-to-SQL fine-tuning under a fixed 500-example budget. It evaluates both an observed-loss IDU adaptation and a training-time gradient IDU adaptation with EXP3 and cumulative replay. All principal methods use five fixed seeds and the official Spider exact-match evaluator.

## Executive summary

Observed-loss IDU achieved the strongest selected-data mean at **17.08%**, compared with 15.64% for Random. The closer-to-LEAD training-time gradient adaptation achieved **15.94%** with EXP3 and cumulative replay. It exceeded Random in three of five seeds, but its paired 95% interval also crossed zero. Neither adaptive result supports a claim of stable superiority.

Full-data training remained clearly stronger at 23.02%. The dynamic gradient method required 489.2 seconds end to end, 29.4% less than Full Data, while losing 7.08 percentage points. The evidence therefore demonstrates a measurable cost-quality trade-off, not full-data parity.

## 1. Research questions

- Q1. Can a 500-example selected subset match or exceed training on all 1,000 candidate examples?
- Q2. Does iterative model-aware selection outperform Random under the same selected-data and optimizer-step budget?
- Q3. Does the iterative method improve the end-to-end performance-cost trade-off?
- Q4. Does iterative selection help relative to a fixed subset, and do gradient IDU and EXP3 provide stable gains?

## 2. Experimental design

| Item                   | Controlled setting                                                                        |
|------------------------|-------------------------------------------------------------------------------------------|
| Dataset                | Spider 1.0; 1,000-example training candidate pool; full 1,034-example development set     |
| Model                  | Salesforce CodeT5-small                                                                   |
| Selected-data budget   | 500 examples for every selected-data method                                               |
| Seeds                  | 11, 42, 73, 101, 202                                                                      |
| Primary metric         | Official Spider exact match                                                               |
| Selected-data training | Five-round methods: 100 examples per round; cumulative union; 1,500 total optimizer steps |
| Cost boundary          | Selection/scoring + training + generation + official evaluation                           |

### Fairness controls

Random and all iterative selected-data methods use the same 500-example budget, model, seeds, development set, and primary metric. The five-round variants also share the same cumulative training schedule and total optimizer-step budget. Full Data uses all 1,000 examples for three epochs; its larger training cost is included in the end-to-end comparison.

### 3. Adaptation of LEAD to Spider

LEAD uses iterative model feedback, Instance-Level Dynamic Uncertainty (IDU), and coarse-to-fine allocation with a multi-armed bandit. This project implements a transparent, computationally manageable Spider adaptation and explicitly separates the paper-aligned components from the task-specific approximations.

- Static uncertainty ranks examples once by pretrained target loss and does not update utility.
- Observed-loss IDU selects 100 examples per round, trains on the cumulative selected set, rescales utility from measured target-loss change, and then selects the next batch.
- Gradient IDU estimates first-order loss change during ordinary training from the final CodeT5 decoder block, applies historical smoothing, and avoids a separate remaining-pool rescaling pass.
- The dynamic method adds loss-quantile difficulty clusters, Spider database task groups, EXP3 allocation from bounded IDU-reduction rewards, and cumulative replay across five rounds.
- A balanced-cluster replay control keeps gradient IDU, grouping, replay, budget, and optimizer steps fixed but replaces EXP3 with the fixed schedule 0, 1, 0, 1, 0.

Gradient update:  $u_i(r) = (1-\beta) \max[0, L_i + g_i^T d\theta] + \beta u_i(r-1)$ . The tracked gradient/update inner product is a first-order proxy for the next loss; cross-entropy is clipped at zero before historical smoothing.

#### Focused hypothesis and competing explanation

Hypothesis: training-time gradient IDU with EXP3 will improve the performance-cost trade-off over Random and a fixed cluster schedule. Competing explanation: any gain may come from cumulative replay and repeated exposure rather than adaptive bandit allocation. The balanced replay control isolates this explanation while preserving the remaining training design.

#### Dynamic gradient LEAD adaptation for Spider Text-to-SQL

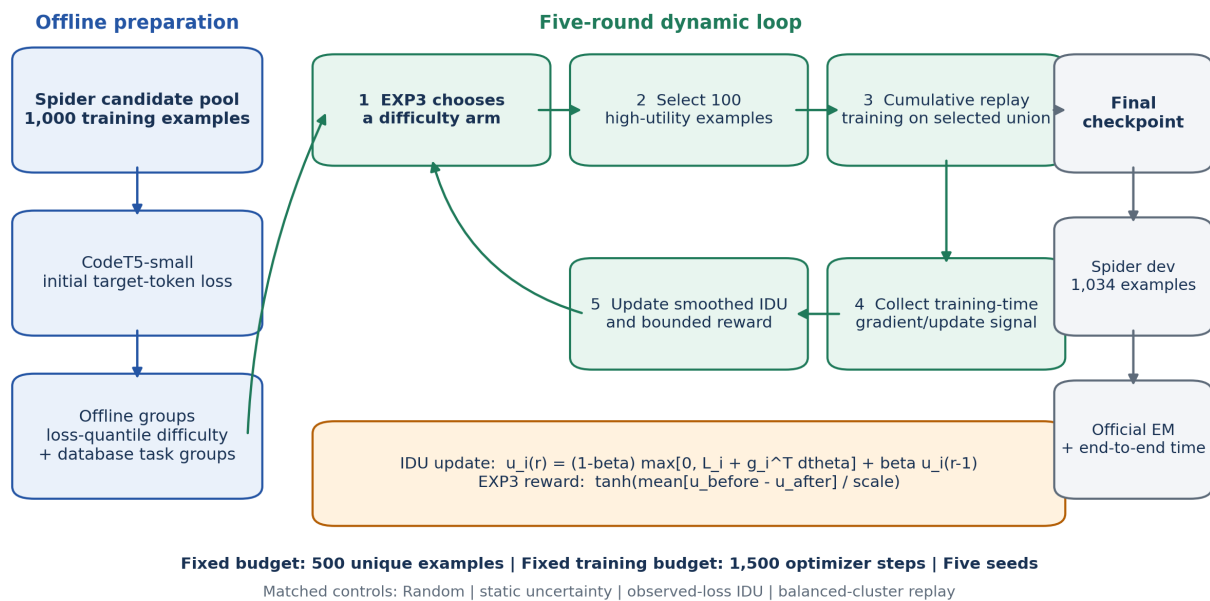


Figure 1. Five-round dynamic gradient LEAD adaptation and matched evaluation boundary.

## 4. Main performance and cost results

| Method                         | Official EM               | End-to-end     | Databases    |
|--------------------------------|---------------------------|----------------|--------------|
| Full Data                      | 23.02% +/- 1.83 pp        | 692.8 s        | 137.0        |
| Random                         | 15.64% +/- 1.13 pp        | 502.4 s        | 126.8        |
| Two-round Iterative            | 13.56% +/- 1.92 pp        | 510.3 s        | 124.0        |
| Five-round Full Adaptation     | 15.22% +/- 2.96 pp        | 605.0 s        | 123.2        |
| <b>Five-round IDU-only</b>     | <b>17.08% +/- 2.39 pp</b> | <b>527.3 s</b> | <b>119.2</b> |
| Static Uncertainty             | 13.82% +/- 2.37 pp        | 457.5 s        | 114.0        |
| Cluster-MAB Component          | 14.24% +/- 1.13 pp        | 629.8 s        | 120.2        |
| Database-quota Component       | 15.52% +/- 2.04 pp        | 582.3 s        | 111.0        |
| <b>Dynamic Gradient + EXP3</b> | <b>15.94% +/- 2.32 pp</b> | <b>489.2 s</b> | <b>123.2</b> |
| Balanced Gradient Control      | 15.02% +/- 3.05 pp        | 430.9 s        | 122.0        |

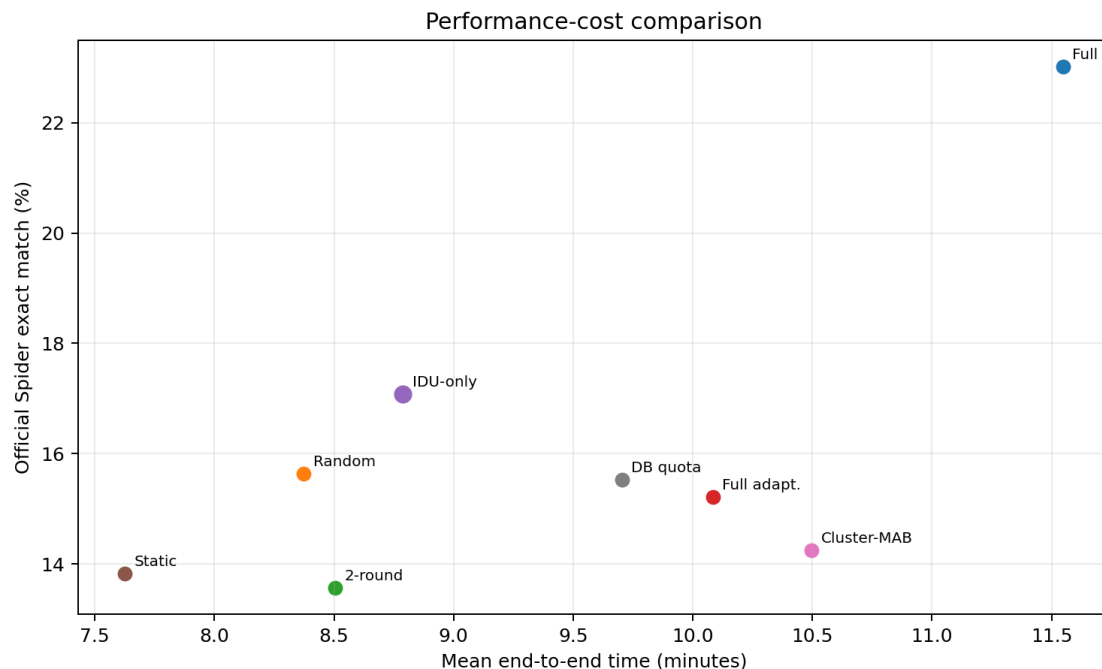


Figure 2. Mean official exact match against measured end-to-end time for the initial comparison set.

### Paired seed analysis

IDU-only minus Random by seed was +1.8, -2.4, +3.5, +5.5, and -1.2 percentage points. The exact sign-flip p-value was 0.4375. With only five seeds, the experiment has low statistical power; the correct conclusion is an observed mean advantage with meaningful seed sensitivity.

### Answers to Q1-Q4

Q1: No. The best selected-data method remains 5.94 percentage points below Full Data.

Q2: Partly. Observed-loss IDU has the highest selected-data mean; dynamic gradient selection is only 0.30 percentage points above Random and is not reliably superior.

Q3: Yes as a trade-off, not as dominance. Dynamic gradient selection is substantially faster than Full Data, but the accuracy gap remains large and its small timing difference from Random should not be overinterpreted.

Q4: Partly. Observed-loss IDU beats the fixed static control, so iterative utility updates can help. However, gradient IDU + EXP3 is only +0.30 pp versus Random and +0.92 pp versus the balanced control, with both paired 95% intervals crossing zero. It changes selection, but does not yield a stable gain.

## 5. Difficulty and SQL-component analysis

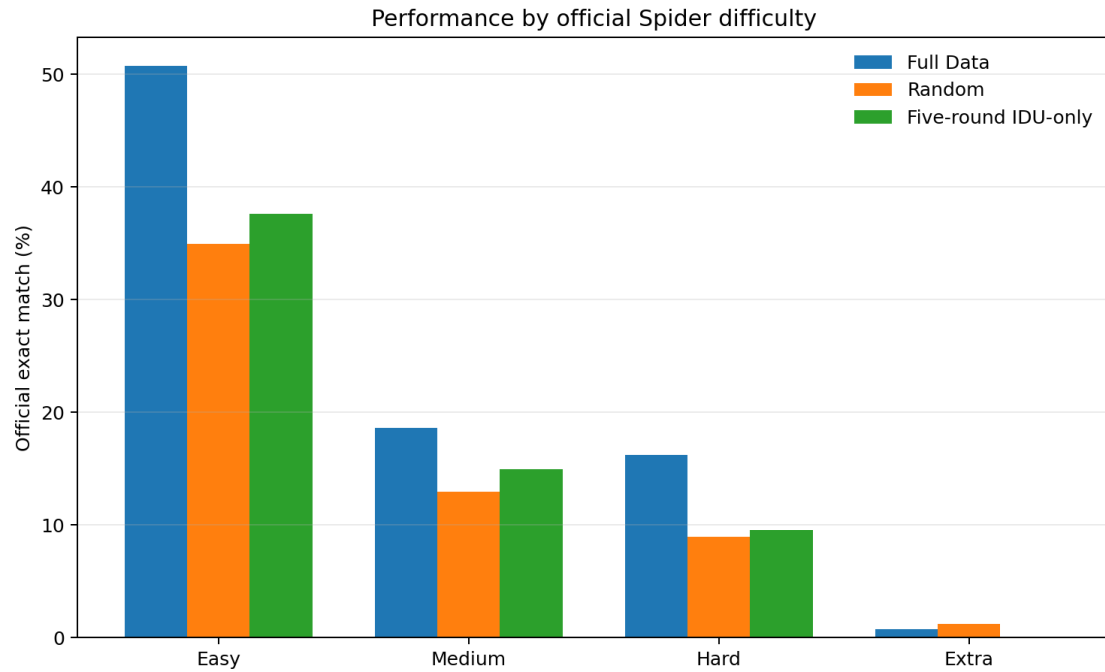


Figure 3. Mean official exact match by Spider difficulty.

| Method              | Easy   | Medium | Hard   | Extra | All    |
|---------------------|--------|--------|--------|-------|--------|
| Full Data           | 50.72% | 18.62% | 16.20% | 0.72% | 23.02% |
| Random              | 34.94% | 12.96% | 8.96%  | 1.20% | 15.64% |
| Five-round IDU-only | 37.64% | 14.94% | 9.54%  | 0.00% | 17.08% |

IDU-only improves on Random for Easy (+2.70 pp), Medium (+1.98 pp), and Hard (+0.58 pp), but does not solve any Extra examples across the five runs. The benefit is therefore concentrated in less structurally demanding queries.

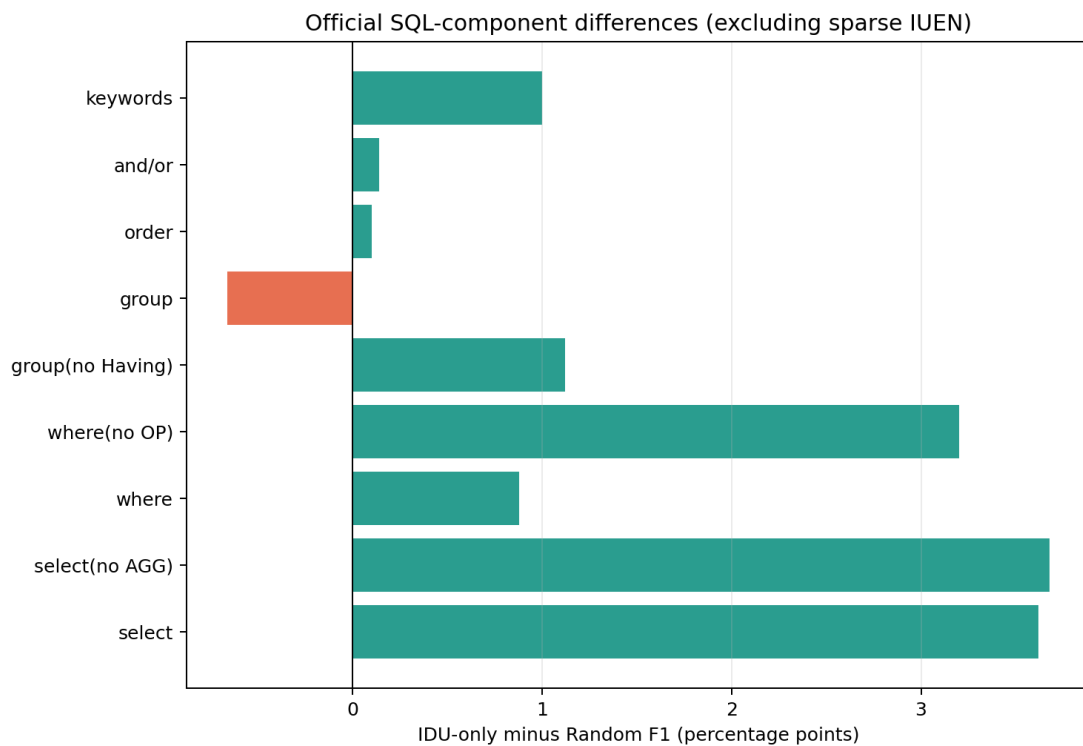


Figure 4. IDU-only minus Random in official partial-match F1; sparse IUEN is excluded from the plot.

The largest stable positive differences are SELECT without aggregation (+3.68 pp), SELECT (+3.62 pp), and WHERE without operator (+3.20 pp). GROUP is slightly lower (-0.66 pp). The official IUEN component is sparse and unstable when a run produces no matching set operation, so its large numerical difference is not interpreted independently.

## 6. What the ablations show

| Comparison                  | Mean difference | Interpretation                                                       |
|-----------------------------|-----------------|----------------------------------------------------------------------|
| IDU-only vs Static          | +3.26 pp        | Online utility updating adds value.                                  |
| IDU-only vs Full Adaptation | +1.86 pp        | All allocation components together do not help.                      |
| IDU-only vs Cluster-MAB     | +2.84 pp        | Current clustering and bandit allocation are too coarse or unstable. |
| IDU-only vs Database quota  | +1.56 pp        | Proportional quotas reduce useful flexibility.                       |

**The central positive result is not that the complete LEAD-style adaptation wins. Instead, the simplest iterative component - updating sample utility from model feedback - is the strongest variant. Adding cluster scheduling or database quotas increases cost without improving the mean.**

### Selection behaviour

IDU-only and Random have a mean selected-set Jaccard overlap of 0.329. IDU-only covers 119.2 databases on average, versus 126.8 for Random. Its observed advantage does not come from maximising database count; it comes from selecting a different, dynamically updated subset.

## 7. Gradient-based LEAD validation

| Comparison              | Mean difference | 95% paired interval  | Wins |
|-------------------------|-----------------|----------------------|------|
| Dynamic vs Balanced     | +0.92 pp        | [-3.30 pp, +5.14 pp] | 3/5  |
| Dynamic vs Random       | +0.30 pp        | [-3.02 pp, +3.62 pp] | 3/5  |
| Dynamic vs Observed IDU | -1.14 pp        | [-6.27 pp, +3.99 pp] | 1/5  |
| Dynamic vs Full Data    | -7.08 pp        | [-9.89 pp, -4.27 pp] | 0/5  |

**Cumulative replay is the clearest successful design change. For seed 42, the gradient method rose from 10.3% without replay to 19.5% with replay. This single-seed contrast motivated the multi-seed validation; it should not be treated as an average effect.**

Across five seeds, dynamic gradient + EXP3 obtained 15.94% +/- 2.32 pp. Its mean was 0.92 points above the matched balanced control and 0.30 points above Random, but both paired intervals crossed zero. It was 1.14 points below observed-loss IDU. The correct interpretation is competitive but unstable performance, not reliable superiority.

Dynamic selections had mean pairwise Jaccard overlap 0.80, compared with 1.00 for the fixed schedule. EXP3 therefore changed the selected path, but five allocation decisions were insufficient to learn a consistently better policy. Rewards also increased mainly with training round, suggesting that round progress and difficulty-arm value were not cleanly separated.

## 8. Manually verified qualitative cases

### IDU-only advantage: grouping and HAVING

For 'List all document ids with at least two paragraphs', IDU-only generated the correct Paragraphs + GROUP BY + HAVING structure in four of five seeds. Random failed in all five seeds, often using the Documents table or replacing HAVING with ORDER BY and LIMIT.

### IDU-only advantage: schema linking and comparison

For countries becoming independent after 1950, IDU-only produced the direct country/IndepYear > 1950 query in four seeds. Random frequently selected the city table, equality, or the wrong column.

## Random advantage: negation

For nationality not equal to Russia, Random preserved the inequality in four seeds. IDU-only failed in all seeds by reversing the condition or generating unnecessary nested NOT IN queries.

## Random advantage: literal preservation

For the episode literal 'A Love of a Lifetime', Random preserved the full literal in four seeds. IDU-only repeatedly shortened the literal or selected an invalid output column.

Formatting-only mismatches, such as quotation style or identifier capitalisation, were excluded from the substantive case analysis. These examples explain behaviour but do not replace the official aggregate metric.

## 9. Limitations

- Five seeds expose instability but are insufficient for a strong statistical superiority claim.
- CodeT5-small has limited capacity and all methods remain weak on Extra queries.
- The gradient proxy tracks only the final decoder block of fully fine-tuned CodeT5-small rather than LoRA final-layer geometry used by the original implementation.
- Difficulty groups use pretrained-loss quantiles rather than LEAD's IFD and semantic K-means pipeline.
- EXP3 receives only five decisions, which limits policy learning and makes its reward sensitive to training-round effects.
- End-to-end timing is measured on one Apple MPS environment and should not be treated as hardware-independent throughput.
- Exact match is strict and does not directly measure execution equivalence; auxiliary normalized-string diagnostics are more formatting-sensitive.

## 10. Conclusion

Under a fixed 500-example budget, observed-loss IDU produced the best selected-data mean at 17.08%, while the more paper-aligned training-time gradient adaptation achieved 15.94%, close to Random at 15.64%. The experiment supports iterative utility updating and cumulative replay as useful design elements, but it does not show a stable accuracy benefit from the present gradient proxy or EXP3 allocation. Full Data remains clearly stronger at 23.02%.

**The most defensible outcome is a controlled positive-and-negative finding: cumulative replay prevents severe forgetting and dynamic selection changes the chosen data, while the current EXP3 reward and coarse difficulty clusters do not yield a reliable generalisation gain. This directly identifies when the LEAD adaptation helps and where its assumptions become weak in small-scale Text-to-SQL.**

## 11. Justified next work

1. Redesign the EXP3 reward to separate difficulty-arm value from the general effect of later training rounds, then test it against the same balanced control.
2. Increase allocation frequency or the number of rounds without changing the total unique-data and optimizer-step budgets, giving the bandit more feedback decisions.
3. Replace loss-quantile clusters with Text-to-SQL-aware groups based on schema and SQL structure; use additional seeds or a larger model only after this targeted design test.

## References

- [1] T. Yu et al. Spider: A Large-Scale Human-Labeled Dataset for Complex and Cross-Domain Semantic Parsing and Text-to-SQL Task. EMNLP, 2018.
- [2] Y. Wang et al. CodeT5: Identifier-aware Unified Pre-trained Encoder-Decoder Models for Code Understanding and Generation. EMNLP, 2021.
- [3] X. Lin et al. LEAD: Iterative Data Selection for Efficient LLM Instruction Tuning. PVLDB 19(3):426-439, 2025. doi:10.14778/3778092.3778103.